

Assembly Guide

Cable-Driven Shoulder Exoskeleton Research Platform

*A high-level build guide. For precise geometry and dimensions refer to the CAD files in **CAD/**. For wiring refer to **Electronics/Simple Schematic**.*

1. 3D Printing

All printable components are in CAD/stl/. Print all parts before beginning assembly.

Component	Qty	Description
Arm Strap Support 1 & 2	1 ea.	Supports the arm bar and serves as attachment point for Velcro straps
Ball Joint	1	Mounts the entire exoskeleton frame to the hip/belt
C-Channel Extension	1	Connects the C-channel frame to the ball joint
Cable Holder	3	Guides the Bowden cable and bungee cord along the length of the frame
Cable Mount	1	Attaches the Bowden cable to the arm bar on either side of the fulcrum
EMG Mount	1	Mount for the MyoWare 2.0 EMG sensor
Elastic Mount	1	Mount for the bungee cord passive elastic element
Encoder Mount	1	Mount for the AS5600 encoder sensor
HX711 Mount	1	Mount for the HX711 load cell amplifier board
IMU Mount	1	Mount for the BNO055 IMU sensor
IR Receiver Mount	1	Mount for the HX1838 IR receiver
Load Cell Mount	4	Connects both halves of the arm bar through the load cell
Servo Mount	1	Mount for the servo motor
Servo Plate	1	Closes the servo gearbox housing
Spacers	Var.	Miscellaneous spacers used throughout the assembly
Standoff Holder	2	Secures the aluminum standoff frame to the C-channel
Winch 1 & 2	1 ea.	Attach to the servo head and drive the Bowden cable; encoder magnet is embedded in the winch

Print settings: 0.2 mm layer height, 20% infill for most parts, 50% infill for load-bearing components (Servo Mount, Load Cell Mount, Arm Strap Supports, Ball Joint).

2. Frame Assembly

1. Attach the Standoff Holders to the aluminum C-channel to form the vertical spine of the frame.
2. Insert aluminum standoffs into the Standoff Holders to connect the spine to the shoulder region.
3. Attach the C-Channel Extension to the bottom of the spine.
4. Mount the Ball Joint onto the C-Channel Extension — this is the primary attachment point to the user's hip/belt.

3. Servo Assembly

1. Seat the servo motor into the Servo Mount and close with the Servo Plate.
2. Attach Winch 1 then Winch 2 to the servo output shaft. The magnet embedded in the winch faces the AS5600 encoder.
3. Mount the Encoder Mount so the AS5600 sensor faces the magnet. Set `ENCODER_OFFSET` in firmware to zero at the neutral arm position.
4. Mount the full servo assembly onto the C-channel frame at shoulder blade height.

4. Bowden Cable and Passive Elastic Routing

1. Thread the inner bicycle cable through the Bowden cable housing.
2. Anchor the sheath at the servo end and route it along the frame, securing at intervals with the Cable Holders.
3. Attach the far end of the inner cable to the arm bar via the Cable Mount on either side of the load cell fulcrum.
4. Attach the bungee cord at both ends using the Elastic Mounts — one on the frame near the shoulder and one on the arm bar. Adjust the anchor position to set passive preload and mark it for consistent re-tensioning between sessions.

5. Arm Bar Assembly

The arm bar runs along the upper arm from the shoulder pivot to just above the elbow. Refer to CAD/ for the full sub-assembly.

1. Assemble the arm bar halves through the four Load Cell Mounts with the load cell seated inline so cable force passes through it.

2. Mount the HX711 board on the arm bar using the HX711 Mount.
3. Attach the two Arm Strap Supports along the arm bar and thread Velcro straps through them to secure the bar to the arm.
4. Mount the IMU Mount near the proximal end of the arm bar and seat the BNO055.
5. Mount the EMG Mount near the deltoid and connect the MyoWare 2.0 sensor.

6. Harness Integration

1. Thread the C-channel spine through the back of the tactical harness and secure with Velcro strips along the spine.
2. Attach the waist strap to the bottom of the assembly.
3. Connect the arm bar to the top of the spine via the shoulder pivot.
4. Mount the IR Receiver Mount on the frame with clear line-of-sight for the remote.
5. Route and bundle all sensor wires along the spine using cable cover sleeving.

7. Electronics

Refer to Electronics/Simple Schematic for all connections.

1. Mount the Arduino Uno on the electronics panel at the waist.
2. Connect the LiPo battery to the Arduino and servo via the 5A inline fuse.
3. Wire all sensors to the Arduino per the schematic.
4. Add the capacitor across the servo power rails.

8. Firmware

1. Install the Arduino IDE and the following libraries via Library Manager: Adafruit BNO055, Adafruit Unified Sensor, Adafruit AS5600, HX711 by bogde, IRremote.
2. Open firmware/exo_firmware.ino and upload to the Arduino Uno.
3. Open Serial Monitor at 115200 baud to verify all sensors are detected.